

	Department of Civil Engineering		
	Even Semester Re-Examination – Academic Year 2023/2024		
	Class: B1C+CE	Subject: Mathematics II	
	Date: / /2024	Time: 2 Hours	MGP:
	Lecturer	Dr. Fadhl Alsarahi	Signature
	Student		Signature

PART (1): Answer the following questions:

Q.(1). Mark each of the following True or False:

1. The equation $\frac{dy}{dx} + P(x)y = Q(x)$, is called Bernoulli's equation. ()
2. Expansion of $\log(1+x)$ is: $1 - x + \frac{x^2}{2} - \frac{x^3}{3} + \dots$. ()
3. If $y = \frac{1}{(ax+b)}$, then $y_n = \frac{(-1)^{n(n-1)!}a^n}{(ax+b)^{n+1}}$. ()
4. If $y = \lim_{x \rightarrow 0} \frac{\cosh x - \cos x}{x \sin x}$, then $y = 1$. ()
5. If $y = e^{ax} \cos(bx + c)$, then $y_n = (a^2 + b^2)^{n/2} e^{ax} \sin\left(bx + c + n \tan^{-1} \frac{b}{a}\right)$. ()

Q.(2). Circle the correct answer in each of the following questions:

- a. If $y = \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2\log(1+x)}{x^2}$, then y equal:

a) $\frac{1}{2}$
b) $\frac{3}{2}$
c) 1
- b. If $y = e^{mx}$, then y_n is equal:

a) $m^n e^{mx} (\log m)^n$
b) $m^n e^{mx}$
c) $n^m e^{mx}$
- c. The differential equation $xy(1 + xy^2)dy = dx$, is:

a) Bernulli's equation
b) exact equation
c) linear equation
- d. If $y = \lim_{x \rightarrow \pi/2} (\sin x)^{\tan x}$, then y is equal:

a) e^{-1}
b) 1
c) 0
- e. The differential equation $M(x, y)dx + N(x, y)dy = 0$, is exact if:

a) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$
b) $\frac{\partial M}{\partial y} = -\frac{\partial N}{\partial x}$
c) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

PART(2): Answer any **THREE** questions from the following questions:

Question(3).

1. Find: $\lim_{x \rightarrow 0} (1 + \sin x)^{\cot x}$
2. Solve Bernoulli's equation: $\frac{dx}{dy} - xy = x^2 y^3$.
3. Solve the exact equation:

$$\left\{ y \left(1 + \frac{1}{x} \right) + \cos y \right\} dx + (x + \log x - x \sin y) dy = 0.$$

Question(4).

- 1) Expansion of $\log(1 - x)$.
- 2) Solve the homogeneous equation

$$x^2 y dx - (x^3 + y^3) dy = 0.$$

- 3) Find: $\lim_{x \rightarrow 0} \frac{x \cos x - \sin x}{x^2 \sin x}$.

Question(5).

- 1) Solve the linear equation $\cos^2 x \frac{dy}{dx} + y = \tan x$.
- 2) Using Maclaurin's series, expand $\frac{\sin^{-1} x}{\sqrt{1-x^2}}$.
- 3) If $y = \cos(m \log x)$, prove that

$$x^2 y_{n+2} + (2n + 1) x y_{n+1} + (n^2 + m^2) y_n = 0$$

Question(6).

1. Solve: $(1 + x^2) \frac{dy}{dx} = (1 + y^2)$.
2. Expansion of $\sqrt{1 + \sin 2x}$.
3. Find the nth derivative of : $\frac{x+3}{(x-1)(x+1)}$.

Good luck